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Probability and Statistics

Subject 5

Several Useful Discrete Distributions

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Contents

- The Binomial Probability Distribution
- The Poisson Probability Distribution
- The Hypergeometric Probability Distribution

Most parts of the slides are derived from the textbook: "Mendenhall, Beaver, Beaver, *Introduction to Probability and Statistics*, 14th Ed., Brooks/Cole, Cengage Learning, 2013"

Introduction

- Discrete random variables take on only a finite or countably infinite number of values.
- Three discrete probability distributions serve as models for a large number of practical applications:

✓The **binomial** random variable

✓The **Poisson** random variable

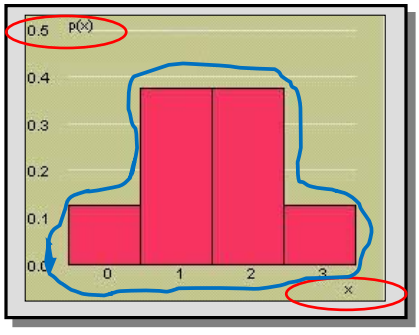
✓The **hypergeometric** random variable

Introduction

- These three discrete random variables are often used to describe the number of occurrences of a specified event in a *fixed number of trials* or *a fixed unit of time or space*.

The Binomial Random Variable

- The **coin-tossing experiment** is a simple example of a **binomial random variable.**
- Toss a fair coin n = 3 times and record x = number of heads.



x	$p(x)$
0	1/8
1	3/8
2	3/8
3	1/8

The Binomial Random Variable

Many situations in real life resemble the coin toss, but the coin is not necessarily fair, so that P(H) ≠ 1/2.

Example: A geneticist samples 10 people and counts the number who have a gene linked to Alzheimer's disease.

- Coin: Person
- Head: Has gene
- Tail: Doesn't have gene
- Number of tosses: n = 10
- P(H): P(has gene) = proportion in the population who have the gene.

The Binomial Experiment

1. The experiment consists of n identical trials.
2. Each trial results in one of two outcomes, success (S) or failure (F).
3. The probability of success on a single trial is p and remains constant from trial to trial. The probability of failure is $q = 1 - p$.
4. The trials are independent.
5. We are interested in x, the number of successes in n trials.

Binomial or Not?

Very few real life applications satisfy these requirements exactly.

- Select two people from the U.S. population, and suppose that 15% of the population has the Alzheimer's gene.
  - For the first person, p = P(gene) = .15
  - For the second person, $p \approx$ P(gene) = .15, even though one person has been removed from the population.

Binomial or Not?

- Keep in mind and check for the below:
  - When the sample (the n identical trials) came from a large population, the probability of success p stayed about the same from trial to trial. Binomial
  - When the population size N was small, the probability of success p changed quite dramatically from trial to trial, and the experiment *was not binomial*.
- Rule of Thumb:
  - If $n/N \geq .05$, the experiment *is not binomial*.

The Binomial Probability Distribution

- For a binomial experiment with n trials and probability p of success on a given trial, the probability of k successes in n trials is

$$P(x=k) = C_k^n p^k q^{n-k} = \frac{n!}{k!(n-k)!} p^k q^{n-k} \text{ for } k = 0,1,2,\ldots n.$$

Recall $C_k^n = \dfrac{n!}{k!(n-k)!}$

with $n! = n(n-1)(n-2)\ldots(2)1$ and $0! \equiv 1$.

The Mean and Standard Deviation

- For a binomial experiment with n trials and probability p of success on a given trial, the measures of center and spread are:

$$\text{Mean}: \mu = np$$
$$\text{Variance}: \sigma^2 = npq$$
$$\text{Standard deviation}: \sigma = \sqrt{npq}$$

Ex. 5.3 pp.179

Example

A marksman hits a target 80% of the time. He fires five shots at the target. What is the probability that exactly 3 shots hit the target?

***n* = 5** ***success* = hit** ***p* = .8** ***x* = # of hits**

$$P(x=3) = C_3^n p^3 q^{n-3} = \frac{5!}{3!2!}(.8)^3(.2)^{5-3}$$

$$= 10(.8)^3(.2)^2 = .2048$$

Example

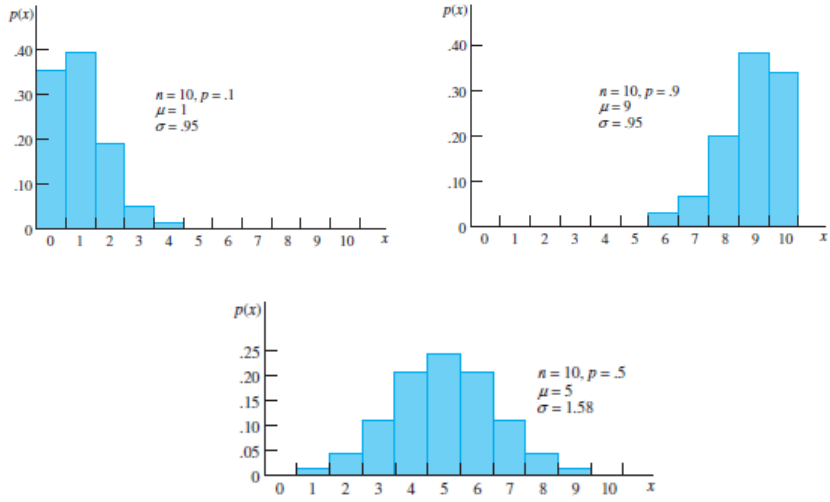
What is the probability that more than 3 shots hit the target?

$$P(x>3) = C_4^5 p^4 q^{5-4} + C_5^5 p^5 q^{5-5}$$

$$= \frac{5!}{4!1!}(.8)^4(.2)^1 + \frac{5!}{5!0!}(.8)^5(.2)^0$$

$$= 5(.8)^4(.2) + (.8)^5 = .7373$$

Example



Cumulative Probability Tables

You can use the **cumulative probability tables** to find probabilities for selected binomial distributions.

1) Find the table for the correct value of ***n.***

2) Find the column for the correct value of ***p.***

3) The row marked "*k*" gives the cumulative probability, $P(x \leq k) = P(x = 0) + \ldots + P(x = k)$

Example

k	*p* = .80
0	.000
1	.007
2	.058
3	.263
4	.672
5	1.000

What is the probability that *exactly 3 shots* hit the target?

P(*x* = 3) = $P(x \leq 3) - P(x = 2) - P(x = 1) - P(x = 0)$
P(*x* = 3) = $P(x \leq 3) - P(x \leq 2)$
= .263 - .058 = .205

Check from formula: P(*x* = 3) = .2048

n = 5

							p							
k	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99	*k*
0	.951	.774	.590	.328	.168	.078	.031	.010	.002	.000	.000	.000	.000	0
1	.999	.977	.919	.737	.528	.337	.188	.087	.031	.007	.000	.000	.000	1
2	1.000	.999	.991	.942	.837	.683	.500	.317	.163	.058	.009	.001	.000	2
3	1.000	1.000	1.000	.993	.969	.913	.812	.663	.472	.263	.081	.023	.001	3
4	1.000	1.000	1.000	1.000	.998	.990	.969	.922	.832	.672	.410	.226	.049	4
5	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	5

Example

k	$p = .80$
0	.000
1	.007
2	.058
3	.263
4	.672
5	1.000

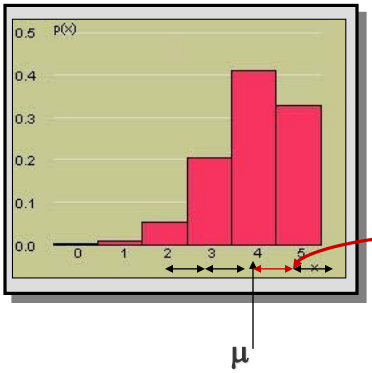
What is the probability that more than 3 shots hit the target?

$$P(x > 3) = 1 - P(x \leq 3)$$
$$= 1 - .263 = .737$$

Check from formula: P(*x* > 3) = .7373

Example

Here is the probability distribution for x = number of hits. What are the mean and standard deviation for x?



$$\text{Mean} : \mu = np = 5(.8) = 4$$

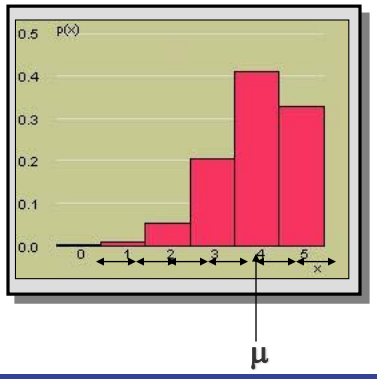
$$\text{Standard deviation} : \sigma = \sqrt{npq}$$

$$= \sqrt{5(.8)(.2)} = .89$$

Example

Would it be unusual to find that none of the shots hit the target?

$$\mu = 4; \sigma = .89$$



The value $x = 0$ lies

$$z = \frac{x - \mu}{\sigma} = \frac{0 - 4}{.89} = -4.49$$

more than 4 standard deviations below the mean. Very unusual.

The Poisson Random Variable

- The Poisson random variable x is a model for data that represent the *number of occurrences* of a specified event in a *given unit of time or space*.

- **Examples:**
  - The number of calls received by a switchboard during a given period of time.
  - The number of machine breakdowns in a day
  - The number of traffic accidents at a given intersection during a given time period.

The Poisson Random Variable

- The only assumption needed to model experiments is that the counts or events occur randomly and independently of one another.

The Poisson Probability Distribution

- x is the number of events that occur in a period of time or space during which an average of μ such events can be expected to occur. The probability of k occurrences of this event is

$$P(x=k)=\frac{\mu^k e^{-\mu}}{k!}$$

For values of k = 0, 1, 2, … The mean and standard deviation of the Poisson random variable are

Mean: μ

Standard deviation: $\sigma=\sqrt{\mu}$

where e = 2.71828…

Example

The average number of traffic accidents on a certain section of highway is two per week. Find the probability of exactly one accident during a one-week period.

Mean, μ = 2 accidents/week

$$P(x=1)=\frac{\mu^k e^{-\mu}}{k!}=\frac{2^1 e^{-2}}{1!}=2e^{-2}=.2707$$

Cumulative Probability Tables

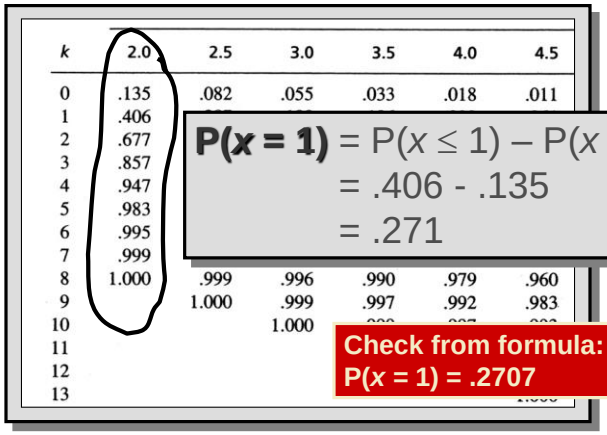
You can use the cumulative probability tables to find probabilities for selected Poisson distributions.

1) Find the column for the correct value of μ.
2) The row marked "k" gives the cumulative probability, $P(x \le k) = P(x = 0) + \ldots + P(x = k)$

Example

k	$\mu = 2$
0	.135
1	.406
2	.677
3	.857
4	.947
5	.983
6	.995
7	.999
8	1.000

What is the probability that there is exactly 1 accident?



k	2.0	2.5	3.0	3.5	4.0	4.5
0	.135	.082	.055	.033	.018	.011
1	.406					
2	.677					
3	.857					
4	.947					
5	.983					
6	.995					
7	.999					
8	1.000	.999	.996	.990	.979	.960
9		1.000	.999	.997	.992	.983
10			1.000			
11						
12						
13						

$$\mathbf{P(x = 1)} = P(x \le 1) - P(x \le 0)$$
$$= .406 - .135$$
$$= .271$$

Check from formula: P(x = 1) = .2707

Example

k	$\mu = 2$
0	.135
1	.406
2	.677
3	.857
4	.947
5	.983
6	.995
7	.999
8	1.000

What is the probability that 8 or more accidents happen?

$$\mathbf{P(x \ge 8)} = 1 - P(x < 8)$$
$$= 1 - P(x \le 7)$$
$$= 1 - .999 = .001$$

This would be very unusual (small probability) since x = 8 lies

$$z = \frac{x - \mu}{\sqrt{\mu}} = \frac{8 - 2}{1.414} = 4.24$$

standard deviations above the mean.

The Poisson Approximation to the Binomial Distribution

- When the number of trial for the Binomial experiment is high, then it is not easy to compute the probabilities and find out the tables for the cumulative probabilities.
- We *can estimate* binomial probabilities with the Poisson when *n* is large and *p* is small.
- The Poisson probability distribution provides a simple, easy-to-compute, and accurate approximation to binomial probabilities when *n is large* and ***μ = np is small***, preferably with ***np < 7***.

The Poisson Approximation to the Binomial Distribution

Example: Suppose a life insurance company insures the lives of 5000 men aged 42. If actuarial studies show the probability that any 42-year-old man will die in a given year to be .001, find the exact probability that the company will have to pay x = 4 claims during a given year.

1st Solution:

$$P(x = 4) = p(4) = \frac{5000!}{4!4996!}(.001)^4(.999)^{4996}$$

2nd Solution (use Poisson Distribution):

$\mu = n.p = 5000 \times 0.001 = 5$

$$p(4) \approx \frac{\mu^4 e^{-\mu}}{4!} = \frac{5^4 e^{-5}}{4!} = \frac{(625)(.006738)}{24} = .175$$

or from Table (Poisson Cumulative Distribution) with μ = 5:

$$p(4) = P(x \leq 4) - P(x \leq 3) = .440 - .265 = .175$$

The Hypergeometric Probability Distribution

- Remember the requirements in a binomial experiment.
- If the number of elements in the population is small in relation to the sample size ($n/N \geq 0.05$), the probability of a success for a given trial is dependent on the outcomes of preceding trials.
- Then the number of x of successes follows what is known as a hypergeometric probability distribution.

The Hypergeometric Probability Distribution

- The "candies problems" from Chapter 4 are modeled by the hypergeometric distribution.
- A bowl contains M red candies and N–M blue candies. Select n candies from the bowl and record x the number of red candies selected. Define a "red candies" to be a "success".

The probability of exactly k successes in n trials is

$$P(x=k)=\frac{C_k^M C_{n-k}^{N-M}}{C_n^N}$$

M → successes, ***N-M*** → failures, ***n*** → size of the random sample space

The Mean and Variance

The mean and variance of the hypergeometric random variable x resemble the mean and variance of the binomial random variable:

$$\text{Mean}: \mu = n\left(\frac{M}{N}\right)$$

$$\text{Variance}: \sigma^2 = n\left(\frac{M}{N}\right)\left(\frac{N-M}{N}\right)\left(\frac{N-n}{N-1}\right)$$

Example

A package of 8 AA batteries contains 2 batteries that are defective. A student randomly selects four batteries and replaces the batteries in his calculator. What is the probability that all four batteries work?

Success = working battery

N = 8

M = 6

***n* = 4**

$$P(x=4) = \frac{C_4^6 C_0^2}{C_4^8}$$

$$= \frac{6(5)/2(1)}{8(7)(6)(5)/4(3)(2)(1)} = \frac{15}{70}$$

Example

What are the mean and variance for the number of batteries that work?

$$\mu = n\left(\frac{M}{N}\right) = 4\left(\frac{6}{8}\right) = 3$$

$$\sigma^2 = n\left(\frac{M}{N}\right)\left(\frac{N-M}{N}\right)\left(\frac{N-n}{N-1}\right)$$
$$= 4\left(\frac{6}{8}\right)\left(\frac{2}{8}\right)\left(\frac{4}{7}\right) = .4286$$

Key Concepts

I. The Binomial Random Variable

1. Five characteristics: *n* identical independent trials, each resulting in either success ***S*** or failure ***F***; probability of success is ***p*** and remains constant from trial to trial; and ***x*** is the number of successes in ***n*** trials.
2. Calculating binomial probabilities
   a. Formula: $P(x = k) = C_k^n p^k q^{n-k}$
   b. Cumulative binomial tables
   c. Individual and cumulative probabilities using Minitab
3. Mean of the binomial random variable: $\mu = np$
4. Variance and standard deviation: $\sigma^2 = npq$ and $\sigma = \sqrt{npq}$

Key Concepts

II. The Poisson Random Variable

1. The number of events that occur in a period of time or space, during which an average of μ such events are expected to occur
2. Calculating Poisson probabilities

   a. Formula: $P(x=k)=\frac{\mu^k e^{-\mu}}{k!}$

   b. Cumulative Poisson tables

   c. Individual and cumulative probabilities using Minitab
3. Mean of the Poisson random variable: $E(x)=\mu$
4. Variance and standard deviation: $\sigma^2=\mu$ and $\sigma=\sqrt{\mu}$
5. Binomial probabilities can be approximated with Poisson probabilities when $np<7$, using $\mu=np$.

Key Concepts

III. The Hypergeometric Random Variable

1. The number of successes in a sample of size *n* from a finite population containing *M* successes and *N* - *M* failures
2. Formula for the probability of *k* successes in *n* trials:

$$P(x=k)=\frac{C_k^M C_{n-k}^{N-M}}{C_n^N}$$

3. Mean of the hypergeometric random variable: $\mu=n\left(\frac{M}{N}\right)$
4. Variance and standard deviation:

$$\sigma^2=n\left(\frac{M}{N}\right)\left(\frac{N-M}{N}\right)\left(\frac{N-n}{N-1}\right)$$